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### System and method with masking for certified defenses against adversarial patches

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#### Abstract

A computer-implemented system and method relate to certified robust defenses against adversarial patches. A set of one-mask images are generated using a source image and a first mask at a set of predetermined image regions. The set of predetermined image regions collectively cover at least every pixel of the source image. A particular one-mask image with a highest prediction loss is selected from among the set of one-mask images. A set of two-mask images is generated using the selected one-mask image and a second mask at the set of predetermined image regions. A particular two-mask image with a highest prediction loss is selected from among the set of two-mask images. The machine learning system is trained using a training dataset, which includes the selected two-mask image.

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## Background/Summary

### TECHNICAL FIELD

(1) This disclosure relates generally to computer vision, and more particularly to digital image processing with certified defenses against adversarial patch attacks.

### BACKGROUND

(2) Deep learning based image classifiers are vulnerable to adversarial patches applied to the input, where image pixels bounded within a confined connected region, usually square or circular, can be adversarial and crafted to induce misclassification. Patch attacks are formulated in this way to mimic printing and placing an adversarial object in the scene. It is easier to realize a patch attack in the physical world compared to full image adversarial perturbation, as the latter might need a compromise in a computing system. Therefore, adversarial patch attacks pose a significant threat to real-world computer vision systems.

### SUMMARY

(3) The following is a summary of certain embodiments described in detail below. The described aspects are presented merely to provide the reader with a brief summary of these certain embodiments and the description of these aspects is not intended to limit the scope of this disclosure. Indeed, this disclosure may encompass a variety of aspects that may not be explicitly set forth below.

(4) According to at least one aspect, a computer-implemented method relates to certified robust defenses against adversarial patches. The method includes receiving a source image. The method includes generating a set of one-mask images using the source image and a first mask at a set of predetermined image regions. Each one-mask image includes the first mask at a different

predetermined image region of the source image. The set of predetermined image regions collectively cover at least every pixel of the source image. The method includes generating, via a machine learning system, a set of predictions in response to receiving the set of one-mask images. The method includes selecting a particular one-mask image with a highest prediction loss from among the set of one-mask images. The method includes generating a set of two-mask images using the selected one-mask image and a second mask at the set of predetermined image regions. Each two-mask image includes the second mask at a different predetermined image region of the selected one-mask image. The method includes generating, via the machine learning system, another set of predictions in response to receiving the set of two-mask images. The method includes selecting a particular two-mask image with a highest prediction loss from among the set of two-mask images. The method includes generating a training dataset that includes the selected two-mask image. The method includes training the machine learning system using the training dataset.

(5) According to at least one aspect, a system includes at least one processor and at least one computer-readable medium. The at least one computer readable medium is in data communication with the at least one processor. The at least one computer-readable medium has computer readable data including instructions stored thereon that, when executed by the at least one processor, causes the at least one processor to perform a method related to certified robust defenses against adversarial patches. The method includes receiving a source image. The method includes generating a set of one-mask images using the source image and a first mask at a set of predetermined image regions. Each one-mask image includes the first mask at a different predetermined image region of the source image. The set of predetermined image regions collectively cover at least every pixel of the source image. The method includes generating, via a machine learning system, a set of predictions in response to receiving the set of one-mask images. The method includes selecting a particular one-mask image with a highest prediction loss from among the set of one-mask images. The method includes generating a set of two-mask images using the selected one-mask image and a second mask at the set of predetermined image regions. Each two-mask image includes the second mask at a different predetermined image region of the selected one-mask image. The method includes generating, via the machine learning system, another set of predictions in response to receiving the set of two-mask images. The method includes selecting a particular two-mask image with a highest prediction loss from among the set of two-mask images. The method includes generating a training dataset that includes the selected two-mask image. The method includes training the machine learning system using the training dataset.

(6) According to at least one aspect, one or more non-transitory computer readable mediums have computer readable data including instructions stored thereon that, when executed by at least one processor, cause the at least one processor to perform a method relating to certified defenses against adversarial patches. The method includes receiving a source image. The method includes generating a set of one-mask images using the source image and a first mask at a set of predetermined image regions. Each one-mask image includes the first mask at a different predetermined image region of the source image. The set of predetermined image regions collectively cover at least every pixel of the source image. The method includes generating, via a machine learning system, a set of predictions in response to receiving the set of one-mask images. The method includes selecting a particular one-mask image with a highest prediction loss from among the set of one-mask images. The method includes generating a set of two-mask images using the selected one-mask image and a second mask at the set of predetermined image regions. Each two-mask image includes the second mask at a different predetermined image region of the selected one-mask image. The method includes generating, via the machine learning system, another set of predictions in response to receiving the set of two-mask images. The method includes selecting a particular two-mask image with a highest prediction loss from among the set of two-mask images. The method includes generating a training dataset that includes the selected two-mask image. The method includes training the machine learning system using the training dataset.

(7) These and other features, aspects, and advantages of the present invention are discussed in the following detailed description in accordance with the accompanying drawings throughout which like characters represent similar or like parts.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram of an example of a system configured to provide certified defenses against adversarial patches according to an example embodiment of this disclosure.
- (2) FIG. 2A is a diagram that shows an example of a set of one-mask images to illustrate each of the predetermined locations of a mask with respect to the source image according to an example embodiment of this disclosure.
- (3) FIG. 2B is a diagram that shows another example of a set of one-mask images to illustrate each of the predetermined locations of a mask with respect to the source image according to an example embodiment of this disclosure.
- (4) FIG. 3 is a diagram that illustrates aspects of the greedy cutout system using a non-limiting example according to an example embodiment of this disclosure.
- (5) FIG. 4 is a diagram of an example of another system that is configured to provide certified defenses against adversarial patches and defend against adversarial patch attacks according to an example embodiment of this disclosure.
- (6) FIG. 5 is a diagram of the system of FIG. 4 with respect to mobile machine technology according to an example embodiment of this disclosure.
- (7) FIG. 6 is a diagram of the system of FIG. 4 with respect to security technology according to an example embodiment of this disclosure.
- (8) FIG. 7 is a diagram of the system of FIG. 4 with respect to imaging technology according to an example embodiment of this disclosure.
- (9) FIG. 8A illustrates non-limiting reference examples of digital images in which mask portions fully occlude the object of interest according to an example embodiment of this disclosure.
- (10) FIG. 8B illustrates non-limiting reference examples of digital images in which mask portions partially occlude the object of interest according to an example embodiment of this disclosure.

### DETAILED DESCRIPTION

- (11) The embodiments described herein, which have been shown and described by way of example, and many of their advantages will be understood by the foregoing description, and it will be apparent that various changes can be made in the form, construction, and arrangement of the components without departing from the disclosed subject matter or without sacrificing one or more of its advantages. Indeed, the described forms of these embodiments are merely explanatory. These embodiments are susceptible to various modifications and alternative forms, and the following claims are intended to encompass and include such changes and not be limited to the particular forms disclosed, but rather to cover all modifications, equivalents, and alternatives falling within the spirit and scope of this disclosure.
- (12) In general, defenses against patch attacks fall into two categories: empirical and certified defense. Empirical defenses often utilize robust training, which incorporates adversarial inputs generated from specific attacks to improve robustness. However, empirical defenses are susceptible to attacks unseen during training. On the other hand, certified defenses guarantee correct predictions against any adaptive white-box attacker under a given threat model. Hence, the certified robust accuracy is the guaranteed lower bound of the model performance.
- (13) FIG. 1 is a diagram of a non-limiting example of a system **100**, which is configured to provide certified defenses against adversarial patch attacks. The system **100** includes a greedy cutout system **130**, which is configured to train a machine learning system **140** with worst-case masked

images. With the greedy cutout system **130**, the system **100** is configured to provide and execute a training strategy, which improves certified robustness over other baselines across different datasets and architectures.

(14) The system **100** includes at least a processing system **110** with one or more processing devices. For example, the processing system **110** includes at least an electronic processor, a central processing unit (CPU), a graphics processing unit (GPU), a microprocessor, a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), any suitable processing technology, or any number and combination thereof. The processing system **110** is operable to provide the functionality as described herein.

(15) The system **100** includes a memory system **120**, which is operatively connected to the processing system **110**. The memory system **120** and the processing system **110** are in data communication with each other. In an example embodiment, the memory system **120** includes at least one non-transitory computer readable storage medium, which is configured to store and provide access to various data to enable at least the processing system **110** to perform the operations and functionality, as disclosed herein. In an example embodiment, the memory system **120** comprises a single memory device or a plurality of memory devices. The memory system **120** may include electrical, electronic, magnetic, optical, semiconductor, electromagnetic, or any suitable storage technology that is operable with the system **100**. For instance, in an example embodiment, the memory system **120** may include random access memory (RAM), read only memory (ROM), flash memory, a disk drive, a memory card, an optical storage device, a magnetic storage device, a memory module, any suitable type of memory device, or any number and combination thereof. With respect to the processing system **110** and/or other components of the system **100**, the memory system **120** is local, remote, or a combination thereof (e.g., partly local and partly remote). For example, the memory system **120** may include at least a cloud-based storage system (e.g. cloud-based database system), which is remote from the processing system **110** and/or other components of the system **100**.

(16) The memory system **120** includes at least a greedy cutout system **130**, a machine learning system **140**, a computer vision application **150**, and other relevant data **160**, which are stored thereon. The greedy cutout system **130** includes computer readable data with instructions, which, when executed by the processing system **110**, is configured to provide certified robust defenses against adversarial patches and/or defend against patch attacks. The computer readable data may include instructions, code, routines, various machine learning data (e.g., parameter data, loss data, training data, etc.), various related data, any software technology, or any number and combination thereof. Also, the machine learning system **140** includes at least one artificial neural network model and/or any suitable machine learning model, which is configured to perform a classification task. In this regard, for example, the machine learning system **140** includes a classifier (e.g., a convolutional neural network (CNN), ResNet, vision transformer (ViT), or any suitable classification model). The machine learning system **140** may be referred to as model  custom character. Also, the computer vision application **150** is configured to apply the output (e.g., class data) of the machine learning system **140** to computer vision technology and/or provide seamless integration of the greedy cutout system **130** and the machine learning system **140** to a downstream application. Meanwhile, the other relevant data **160** provides various data (e.g. operating system, etc.), which enables the system **100** to perform the functions as discussed herein.

(17) The system **100** is configured to include at least one sensor system **170**. The sensor system **170** includes one or more sensors. For example, the sensor system **170** includes an image sensor, a camera, a radar sensor, a light detection and ranging (LIDAR) sensor, a thermal sensor, an ultrasonic sensor, an infrared sensor, a motion sensor, an audio sensor (e.g., microphone), any suitable sensor, or any number and combination thereof. The sensor system **170** is operable to communicate with one or more other components (e.g., processing system **110** and memory system **120**) of the system **100**. For example, the sensor system **170** may provide sensor data, which is then

used by the processing system **110** to generate digital image data based on the sensor data. In this regard, the processing system **110** is configured to obtain the sensor data as digital image data directly or indirectly from one or more sensors of the sensor system **170**. The sensor system **170** is local, remote, or a combination thereof (e.g., partly local and partly remote). Upon receiving the sensor data, the processing system **110** is configured to process this sensor data (e.g. image data) in connection with the greedy cutout system **130**, the machine learning system **140**, the computer vision application **150**, the other relevant data **160**, or any number and combination thereof.

(18) In addition, the system **100** may include at least one other component. For example, as shown in FIG. 1, the memory system **120** is also configured to store other relevant data **160**, which relates to operation of the system **100** in relation to one or more components (e.g., sensor system **170**, I/O devices **180**, and other functional modules **190**). In addition, the system **100** is configured to include one or more DO devices **180** (e.g., display device, keyboard device, speaker device, etc.), which relate to the system **100**. Also, the system **100** includes other functional modules **190**, such as any appropriate hardware, software, or combination thereof that assist with or contribute to the functioning of the system **100**. For example, the other functional modules **190** include communication technology (e.g. wired communication technology, wireless communication technology, or a combination thereof) that enables components of the system **100** to communicate with each other as described herein. In this regard, the system **100** is operable to execute the greedy cutout system **130** to defend against adversarial patch attacks with respect to the machine learning system **140**, as described herein.

(19) FIG. 2A and FIG. 2B illustrate two different examples of sets of one-mask images to illustrate each of the predetermined locations of a mask with respect to the source image according to an example embodiment. In FIG. 2A and FIG. 2B, the light gray portions represent image regions and dark gray portions represent mask regions, as indicated by the legend. As shown in FIG. 2A and FIG. 2B, the mask regions collectively satisfy the R-covering property, i.e., for any adversarial patch, there exists at least one mask within the mask set that would completely cover the adversarial patch. In this regard, the system **100** uses datasets, which include at least source images. For instance, regarding datasets of high-resolution images such as ImageNet and ImageNette, the greedy cutout system **130** resizes and crops the images to 224×224 before feeding them into the machine learning system **140**, which includes model  custom character. For datasets with images with low resolution such as CIFAR-10, the greedy cutout system **130** resizes them to 224×224 via bicubic interpolation for a better classification performance. Although the greedy cutout system **130** uses source images, which are “clean” images (e.g. digital images without adversarial pixels), the greedy cutout system **130** is configured to consider a single adversarial square patch applied on a digital image to be a threat. The greedy cutout system **130** provides mask sets (e.g. a mask set M.sub.3×3, a mask set M.sub.6×6, etc.), which ensure that a putative adversarial patch of an estimated size (e.g. 3% pixels being associated with 39×39 patch) will be covered and therefore neutralized by at least one of the masks (e.g., a mask at one of the predetermined image regions).

(20) To raise the certified robust accuracy, the model  custom character is configured to be fine-tuned with respect to two-masked images  $\{x \odot m_{\text{sub.1}} \odot m_{\text{sub.2}} | m_{\text{sub.1}}, m_{\text{sub.2}} \in M \times M\}$  so it is less sensitive to presence of  $m \in M$ . For this purpose, the greedy cutout system **130** uses augmented training data with  $(x', y)$  where  $x'$  is obtain from applying masks or combination of masks on  $x$ . In this regard, the greedy cutout system **130** develops a systematic approach to find  $x'$  that are more empirically effective than other approaches, such as PatchCleanser. The greedy cutout system **130** uses several inference rounds to identify worst-case masks that lead to highest losses (e.g., highest prediction/classification losses). Then, the greedy cutout system **130** applies these worst- case masks to  $x$  to obtain augmented data point  $(x', y)$  for training the machine learning system **140**.

(21) FIG. 2A shows an example of M.sub.3×3, which is a set of 3×3 masks with nine masks in total, and where each mask is of size 100×100. Specifically, FIG. 2A illustrates a set **200** of one-mask images. Each one-mask image includes a mask at a different predetermined location of the

digital image. For example, the set **200** of one-mask images includes (1) a one-mask image **200A** with a mask at a first predetermined image region of the digital image; (2) a one-mask image **200B** with a mask at a second predetermined image region of the digital image; (3) a one-mask image **200C** with a mask at a third predetermined image region of the digital image; (4) a one-mask image **200D** with a mask at a fourth predetermined image region of the digital image; (5) a one-mask image **200E** with a mask at a fifth predetermined image region of the digital image; (6) a one-mask image **200F** with a mask at a sixth predetermined image region of the digital image; (7) a one-mask image **200G** with a mask at a seventh predetermined image region of the digital image; (8) a one-mask image **200H** with a mask at an eighth predetermined image region of the digital image; and (9) a one-mask image **200I** with a mask at a ninth predetermined image region of the digital image. As shown in FIG. 2A, the set **200** includes a set of mask regions, which collectively cover every pixel of the source image (or the digital image).

(22) FIG. 2B shows an example of  $M.sub.6 \times 6$ , which is a set of  $6 \times 6$  masks with thirty-six masks, and every mask is in shape of  $69 \times 69$ . Specifically, FIG. 2B illustrates a set **210** of one-mask images. Each one-mask image includes a mask at a different predetermined location of the digital image. For example, the set **210** of one-mask images comprises (1) a subset **210A** that includes four different one-mask images, (2) a subset **210B** that includes four different one-mask images, (3) a subset **210C** that includes four different one-mask images, (4) a subset **210D** that includes four different one-mask images, (5) a subset **210E** that includes four different one-mask images, (6) a subset **210F** that includes four different one-mask images, (7) a subset **210G** that includes four different one-mask images, (8) a subset **210H** that includes four different one-mask images, and (9) a subset **210I** that includes four different one-mask images. For convenience, each one-mask image is not labelled with a reference number in FIG. 2B, but only some of the one-mask images are labelled so that the labels may be easily viewed without clutter of other reference labels. For example, the subset **210A** is labeled as an illustrative example. For instance, the subset **210A** includes (i) a one-mask image **212** with a mask at a predetermined image region of the digital image that corresponds to a first part of the mask of the one-mask image **200A**, (ii) a one-mask image **214** with a mask at a predetermined image region of the digital image that corresponds to a second part of the mask of the one-mask image **200A**, (iii) a one-mask image **216** with a mask at a predetermined image region of the digital image that corresponds to a third part of the mask region of one-mask image **200A**, and (iv) a one-mask image **218** with a mask at a predetermined image region of the digital image that corresponds to a fourth part of the mask region of one-mask image **200A**. In this regard, the mask regions of the subset **210A** collectively correspond to the single mask region at the first predetermined image region of the one-mask image **200A**. Also, as another example, the subset **210D** includes (i) a one-mask image **220** with a mask at a predetermined image region of the digital image that corresponds to a first part (upper left portion) of the mask of one-mask image **200D**, (ii) a one-mask image **222** with a mask at a predetermined image region of the digital image that corresponds to a second part (upper right portion) of the mask of one-mask image **200D**, (iii) a one-mask image **224** with a mask at a predetermined image region of the digital image that corresponds to a third part (lower left portion) of the mask of one-mask image **200D**, and (iv) a one-mask image **226** with a mask at a predetermined image region of the digital image that corresponds to a fourth part (lower right portion) of the mask of one-mask image **200D**. The mask regions of the subset **210D** collectively correspond to the single mask region at the fourth predetermined image region of the one-mask image **200D**.

(23) Furthermore, due to the careful choice of  $M.sub.3 \times 3$  and  $M.sub.6 \times 6$ , each mask in  $M.sub.3 \times 3$  may be fully covered by a corresponding subset of 4 masks in  $M.sub.6 \times 6$ . For example, FIG. 2B illustrates that (1) a subset **210A** includes four masks in which those four masks correspond to and fully cover a single mask of the one-mask image **200A**, (2) a subset **210B** includes four masks in which those four masks correspond to and fully cover a single mask of the one-mask image **200B**, (3) a subset **210C** includes four masks in which those four masks correspond to and fully cover a



single mask of the one-mask image **200C**, (4) a subset **210D** includes four masks in which those four masks correspond to and fully cover a single mask of the one-mask image **200D**, (5) a subset **210E** includes four masks in which those four masks correspond to and fully cover a single mask of the one-mask image **200E**, (6) a subset **210F** includes four masks in which those four masks correspond to and fully cover a single mask of the one-mask image **200F**, (7) a subset **210G** includes four masks in which those four masks correspond to and fully cover a single mask of the one-mask image **200G**, (8) a subset **210H** includes four masks in which those four masks correspond to and fully cover a single mask of the one-mask image **200H**, and (9) a subset **210I** includes four masks in which those four masks correspond to and fully cover a single mask of the one-mask image **200I**.

(24) Referring to FIG. 2A and FIG. 2B, empirically,  $M_{sub.6 \times 6}$  leads to higher certified robust accuracy since two smaller masks have a smaller effect on the prediction of model  custom character than the larger masks. However,  $M_{sub.6 \times 6}$  has  $\approx 8$  times larger size than  $M_{sub.3 \times 3}$  and requires more computation in both inference and certification. In this regard, the greedy cutout system **130** is configured to strike a delicate trade-off between computational cost and accuracy.

(25) FIG. 3 illustrates aspects of the greedy cutout system **130** via a non-limiting example according to an example embodiment. As aforementioned, the greedy cutout system **130** may be executed by one or more processors of the processing system **110**. The greedy cutout system **130** first defines the prediction loss of the model  custom character on image label pair  $(x, y)$ . For example, suppose within the model  custom character, for each label  $y' \in Y$ , the confidence level for model  custom character to predict  $y'$  is  $p_{sub.y}$ , then the greedy cutout system **130** uses the cross entropy loss  $\text{loss}_{sub.y}(\text{model}(x, y)) = -\sum_{y' \in Y} c_{sub.y'} \log p_{sub.y'}$ , where  $c_{sub.y'} = 1$  for  $y' = y$  otherwise  $c_{sub.y'} = 0$ . For mask set  $M_{sub.k \times k}$  ( $k=3$  or  $k=6$ ), the greedy cutout system **130** is configured to identify the worst-case mask combination  $m_{sub.1}$ ,  $m_{sub.2} \in M_{sub.k \times k} \times M_{sub.k \times k}$  that incurs the largest loss  $\text{loss}_{sub.y}(\text{model}(x \odot m_{sub.1} \odot m_{sub.2}, y))$  with grid search, but inferencing over all

$$(26) \binom{k^2}{2} + k^2$$

(45 when  $k=3$ , and 666 when  $k=6$ ) unique combinations would be computationally prohibitive, thereby motivating the strategy of the greedy cutout system **130**.

(27) The greedy cutout system **130** finds the worst-case mask in each individual masking round. For example, in the first round, for each  $m_{sub.1} \in M_{sub.k \times k}$ , the greedy cutout system **130** computes the loss  $\text{loss}_{sub.y}(\text{model}(x \odot m_{sub.1}, y))$ , and denotes the mask incurring the highest loss as  $m_{sub.1}^{sup.*}$ . Then, in the second round, for each  $m_{sub.2} \in M_{sub.k \times k}$ ;  $m_{sub.2} \neq m_{sub.1}$ , the greedy cutout system **130** computes the loss  $\text{loss}_{sub.y}(\text{model}(x \odot m_{sub.1}^{sup.*} \odot m_{sub.2}, y))$ , and finds the  $m_{sub.2}^{sup.*}$  with the highest loss. Then, the greedy cutout system **130** empirically uses  $M_{sub.2} := m_{sub.1}^{sup.*} \odot m_{sub.2}^{sup.*}$  as the worst-case mask combination. Although  $M_{sub.2}$  may not be the exact worst-case mask,  $(x \odot M_{sub.2}, y)$  still provides a guidance for fine-tuning. More importantly, the greedy cutout system **130** significantly reduces the inference burden down to  $2k_{sup.2} - 1$  (17,  $k=3$ ,  $k=6$ ) compared to grid search. Also, for  $k=6$ , although the computation cost is high, the mask set  $M_{sub.6 \times 6}$  has better accuracy over the mask set  $M_{sub.3 \times 3}$ .

(28) Referring to FIG. 3, as a non-limiting example, in a first round (or “round 1”) of masking, the greedy cutout system **130** applies a mask set to a source image **300**. For example, the mask set may be a mask set  $M_{sub.3 \times 3}$  (FIG. 2A), a mask set  $M_{sub.6 \times 6}$  (FIG. 2B), etc. The source image **300** is a digital image. The source image **300** is a clean image (or an image without any adversarial pixels/patches). For instance, in FIG. 3, when applying  $M_{sub.3 \times 3}$  as the mask set to the source image **300**, the greedy cutout system **130** generates a set **310** of one-mask images using the source



image **300** and a first mask at a set of predetermined image regions of the source image **300**. For example, as shown in FIG. 3, the set **310** of one-mask images include nine one-mask images, which include (1) a one-mask image **310A** with a first mask at a first predetermined image region of the source image **300**, (2) a one-mask image **310B** with the first mask at a second predetermined image region of the source image **300**, (3) a one-mask image **310C** with the first mask at a third predetermined image region of the source image **300**, (4) a one-mask image **310D** with the first mask at a fourth predetermined image region of the source image **300**, (5) a one-mask image **310E** with the first mask at a fifth predetermined image region of the source image **300**, (6) a one-mask image **310F** with the first mask at a sixth predetermined image region of the source image **300**, (7) a one-mask image **310G** with the first mask at a seventh predetermined image region of the source image **300**, (8) a one-mask image **310H** with the first mask at an eighth predetermined image region of the source image **300**, (9) a one-mask image **310I** with the first mask at a ninth predetermined image region of the source image **300**.

(29) As shown in FIG. 3, the first mask (aka “mask 1”) covers every pixel of the source image **300** across the set **310** of one-mask images. For example, when the first mask is located at a first predetermined image region of the source image **300**, the first mask covers a left section of an upper row (e.g., left top corner) of the source image **300**, as shown by the one-mask image **310A**. When the first mask is located at a second predetermined image region of the source image **300**, the first mask covers a center section of an upper row of the source image **300**, as shown by the one-mask image **310B**. When the first mask is located at a third predetermined image region of the source image **300**, the first mask covers a right section of an upper row (e.g., right top corner) of the source image **300**, as shown by the one-mask image **310C**. When the first mask is located at a fourth predetermined image region of the source image **300**, the first mask covers a left section of a middle row of the source image **300**, as shown by the one-mask image **310D**. When the first mask is located at a fifth predetermined image region of the source image **300**, the first mask covers a center section of a middle row of the source image **300**, as shown by the one-mask image **310E**. When the first mask is located at a sixth predetermined image region of the source image **300**, the first mask covers a right section of a middle row of the source image **300**, as shown by the one-mask image **310F**. When the first mask is located at a seventh predetermined image region of the source image **300**, the first mask covers a left section of a lower row (e.g., left bottom corner) of the source image **300**, as shown by the one-mask image **310G**. When the first mask is located at an eighth predetermined image region of the source image **300**, the first mask covers a center section of a lower row of the source image **300**, as shown by the one-mask image **310H**. When the first mask is located at a ninth predetermined image region of the source image **300**, the first mask covers a right section of a lower row (e.g., bottom right corner) of the source image **300**, as shown by the one-mask image **310I**.

(30) As discussed earlier, upon generating the set **310** of one-mask images, the greedy cutout system **130** computes the loss custom character((custom characterx $\otimes$ m.sub.1), y), and denotes the mask incurring the highest loss as m.sub.1.sup.\*. In the non-limiting example of FIG. 3, for instance, the greedy cutout system **130** generates prediction data indicating that the highest loss is associated with the one-mask image **310D** with the first mask at a fourth predetermined image region of the source image **300**. The greedy cutout system **130** selects the one-mask image (e.g., one-mask image **310D**) that incurs the highest classification loss from among the set **310** of one-mask images. The greedy cutout system **130** then uses only the selected one-mask image **310D** in a second round (“round 2”) of masking.

(31) In the second round of masking, the greedy cutout system **130** applies a second mask of a same type of mask set (e.g., M.sub.3 $\times$ 3, M.sub.6 $\times$ 6, etc.) to the selected one-mask image **310D**. In this example, a mask region of the first mask is equivalent or similar to a mask region of the second mask. In FIG. 3, for instance, the mask set M.sub.3 $\times$ 3 is applied in the first round of masking and the second round of masking. In the second round, upon applying M.sub.3 $\times$ 3 as the mask set to the

selected one-mask image **310D**, the greedy cutout system **130** generates a set **320** of two-mask images with (i) the first mask being fixed at its predetermined location of the source image **300** and (ii) the second mask being at a set of predetermined locations of the source image **300**. For example, as shown in FIG. 3, the set **320** of two-mask images include nine two-mask images, which include (1) a two-mask image **320A** with the second mask at a first predetermined image region of the selected one-mask image **310D**, (2) a two-mask image **320B** with the second mask at a second predetermined image region of the selected one-mask image **310D**, (3) a two-mask image **320C** with the second mask at a third predetermined image region of the selected one-mask image **310D**, (4) a two-mask image **320D** with the second mask at a fourth predetermined image region of the selected one-mask image **310D**, (5) a two-mask image **320E** with the second mask at a fifth predetermined image region of the selected one-mask image **310D**, (6) a two-mask image **320F** with the second mask at a sixth predetermined image region of the selected one-mask image **310D**, (7) a two-mask image **320G** with the second mask at a seventh predetermined image region of the selected one-mask image **310D**, (8) a two-mask image **320H** with the second mask at an eighth predetermined image region of the selected one-mask image **310D**, (9) a two-mask image **320I** with the second mask at a ninth predetermined image region of the selected one-mask image **310D**.

(32) As shown in FIG. 3, the second mask covers every pixel of the source image **300** across the set **320** of two-mask images. For example, when the second mask is located at a first predetermined image region of the selected one-mask image **310D**, the second mask covers a left section of an upper row (e.g. top left corner) of the source image **300**, as shown by the two-mask image **320A**. When the second mask is located at a second predetermined image region of the selected one-mask image **310D**, the second mask covers a center section of an upper row of the source image **300**, as shown by the two-mask image **320B**. When the second mask is located at a third predetermined image region of the selected one-mask image **310D**, the second mask covers a right section of an upper row (e.g., top right corner) of the source image **300**, as shown by the two-mask image **320C**. When the second mask is located at a fourth predetermined image region of the selected one-mask image **310D**, the second mask covers a left section of a middle row of the source image **300**, as shown by the two-mask image **320D**. Moreover, as shown in FIG. 3, the two-mask image **320D** illustrates a case in which the first mask and the second mask overlap each other and cover the same fourth predetermined image region. When the second mask is located at a fifth predetermined image region of the selected one-mask image **310D**, the second mask covers a center section of a middle row of the source image **300**, as shown by the two-mask image **320E**. When the second mask is located at a sixth predetermined image region of the selected one-mask image **310D**, the second mask covers a right section of a middle row of the source image **300**, as shown by the two-mask image **320F**. When the second mask is located at a seventh predetermined image region of the selected one-mask image **310D**, the second mask covers a left section of a lower row (e.g., bottom left corner) of the source image **300**, as shown by the two-mask image **320G**. When the second mask is located at an eighth predetermined image region of the selected one-mask image **310D**, the second mask covers a center section of a lower row of the source image **300**, as shown by the two-mask image **320H**. When the second mask is located at a ninth predetermined image region of the selected one-mask image **310D**, the second mask covers a right section of a lower row (e.g., a bottom right corner) of the source image **300**, as shown by the two-mask image **320I**.

(33) As discussed earlier, upon generating the set **320** of two-mask images, the greedy cutout system **130** computes the loss  $\text{custom\_character}(\text{custom\_character}(x_{\text{m.sub.1.sup.*}}, y_{\text{m.sub.2.sup.*}}))$ , and finds the  $\text{m.sub.2.sup.*}$  with the highest loss. Then, the greedy cutout system **130** empirically uses  $\text{M.sub.2} := \text{m.sub.1.sup.*} \odot \text{m.sub.2.sup.*}$  as the worst-case mask combination. In the non-limiting example of FIG. 3, for instance, the greedy cutout system **130** generates prediction data indicating that the highest loss is associated with the two-mask image **320A** with the second mask at a first predetermined image region of the selected one-mask image **310D**. The greedy cutout system **130** selects the two-mask image (e.g., two-mask image **320A**) that incurs the highest

classification loss from among the set **320** of two-mask images. The greedy cutout system **130** then uses only (i) each source image and (ii) each selected two-mask image **320A** in a training dataset. In this regard, the machine learning system **140** is therefore trained on this training dataset, which includes only (i) source images **300** and (ii) worst case two-mask images **320A** (e.g., two-mask images with highest classification losses from first and second rounds of masking).

(34) FIG. **4** is a diagram of a system **400**, which includes the greedy cutout system **130**. The system **400** is configured to also include at least a sensor system **410**, a control system **420**, and an actuator system **430**. The system **400** is configured such that the control system **420** controls the actuator system **430** based on sensor data from the sensor system **410**. More specifically, the sensor system **410** includes one or more sensors and/or corresponding devices to generate sensor data. For example, the sensor system **410** includes an image sensor, a camera, a radar sensor, a light detection and ranging (LIDAR) sensor, a thermal sensor, an ultrasonic sensor, an infrared sensor, a motion sensor, a satellite-based navigation sensor (e.g., Global Positioning System (GPS) sensor), an optical sensor, an audio sensor, any suitable sensor, or any number and combination thereof. Upon obtaining detections from the environment, the sensor system **410** is operable to communicate with the control system **420** via an input/output (I/O) system **470** and/or other functional modules **450**, which includes communication technology.

(35) The control system **420** is configured to obtain the sensor data directly or indirectly from one or more sensors of the sensor system **410**. In this regard, the sensor data may include sensor data from a single sensor or sensor-fusion data from a plurality of sensors. Upon receiving input, which includes at least sensor data, the control system **420** is operable to process the sensor data via the processing system **440**. In this regard, the processing system **440** includes at least one processor. For example, the processing system **440** includes an electronic processor, a central processing unit (CPU), a graphics processing unit (GPU), a microprocessor, a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), processing circuits, any suitable processing technology, or any combination thereof. Upon processing at least this sensor data, the processing system **440** is configured to extract, generate, and/or obtain proper input data (e.g., digital image data) for the greedy cutout system **130**. In addition, the processing system **440** is operable to generate output data (e.g., class data) via the greedy cutout system **130** and the machine learning system **140** based on communications with the memory system **460**. In addition, the processing system **440** is operable to provide actuator control data to the actuator system **430** based on the output data (e.g., class data), which is generated via the greedy cutout system **130** and the machine learning system **140**.

(36) The memory system **460** is a computer or electronic storage system, which is configured to store and provide access to various data to enable at least the operations and functionality, as disclosed herein. The memory system **460** comprises a single device or a plurality of devices. The memory system **460** includes electrical, electronic, magnetic, optical, semiconductor, electromagnetic, any suitable memory technology, or any combination thereof. For instance, the memory system **460** may include random access memory (RAM), read only memory (ROM), flash memory, a disk drive, a memory card, an optical storage device, a magnetic storage device, a memory module, any suitable type of memory device, or any number and combination thereof. In an example embodiment, with respect to the control system **420** and/or processing system **440**, the memory system **460** is local, remote, or a combination thereof (e.g., partly local and partly remote). For example, the memory system **460** may include at least a cloud-based storage system (e.g., cloud-based database system), which is remote from the processing system **440** and/or other components of the control system **420**.

(37) The memory system **460** includes at least the greedy cutout system **130**, which is executed via the processing system **440**. The greedy cutout system **130** is configured to receive or obtain input data, which includes at least one digital image. In this regard, the greedy cutout system **130**, via the machine learning system **140** and the processing system **440**, is configured to generate output data

(e.g., a set of class data) based on the input data (e.g., one or more digital images). The machine learning system **140** includes at least one artificial neural network model and/or any suitable machine learning model, which is configured to perform a classification task. In this regard, for example, the machine learning system **140** includes a classifier (e.g., a convolutional neural network (CNN), ResNet, vision transformer (ViT), or any suitable classification model). The machine learning system **140** may be referred to as model  custom character. In this regard, the greedy cutout system **130** is advantageously configured to implement a first round of masking to generate one-mask images for the digital image. These one-mask images are then input to the machine learning system **140** to generate class data for that digital image by using a majority prediction and disagreeer prediction technique similarly to PatchCleanser. In addition, if deemed necessary, the greedy cutout system **130** is configured to implement a second round of masking to generate two-mask images based on each disagreeing one-mask image of the first round of masking. These two-mask images are then input to the machine learning system **140** to generate class data for the digital image by using a majority prediction and disagreeer prediction technique similarly to PatchCleanser. Also, the memory system **460** includes the computer vision application **150**, which is configured to apply the output (e.g., class data) of the machine learning system **140** to computer vision technology and/or provide seamless integration of the greedy cutout system **130** and the machine learning system **140** to a downstream application. Meanwhile, the other relevant data **480** provides various data (e.g. operating system, etc.), which enables the system **400** to perform the functions as discussed herein.

(38) Furthermore, as shown in FIG. 4, the system **400** includes other components that contribute to operation of the control system **420** in relation to the sensor system **410** and the actuator system **430**. For example, as shown in FIG. 4, the memory system **460** is also configured to store other relevant data **480**, which relates to the operation of the system **400** in relation to one or more components (e.g., sensor system **410**, the actuator system **430**, etc.). Also, as shown in FIG. 4, the control system **420** includes the I/O system **470**, which includes one or more interfaces for one or more I/O devices that relate to the system **400**. For example, the I/O system **470** provides at least one interface to the sensor system **410** and at least one interface to the actuator system **430**. Also, the control system **420** is configured to provide other functional modules **450**, such as any appropriate hardware technology, software technology, or any combination thereof that assist with and/or contribute to the functioning of the system **400**. For example, the other functional modules **450** include an operating system and communication technology that enables components of the system **400** to communicate with each other as described herein. With at least the configuration discussed in the example of FIG. 4, the system **400** is applicable in various technologies.

(39) FIG. 5 is a diagram of the system **400** with respect to mobile machine technology **500** according to an example embodiment. As a non-limiting example, the mobile machine technology **500** includes at least a partially autonomous vehicle or robot. In FIG. 5, the mobile machine technology **500** is at least a partially autonomous vehicle, which includes a sensor system **410**. The sensor system **410** includes an optical sensor, an image sensor, a video sensor, an ultrasonic sensor, a position sensor (e.g. GPS sensor), a radar sensor, a LIDAR sensor, any suitable sensor, or any number and combination thereof. One or more of the sensors may be integrated with respect to the vehicle. The sensor system **410** is configured to provide sensor data to the control system **420**.

(40) The control system **420** is configured to obtain image data, which is based on sensor data or sensor-fusion data from the sensor system **410**. In addition, the control system **420** is configured to pre-process the sensor data to provide input data of a suitable form (e.g., digital image data) to the greedy cutout system **130** and the machine learning system **140**. With the guidance of the greedy cutout system **130**, the machine learning system **140** generates class data as output upon receiving a given digital image as input with certified robust defenses against adversarial patches.

(41) Upon receiving class data from the machine learning system **140**, the greedy cutout system **130**, and/or the computer vision application **150**, the control system **420** is configured to generate

actuator control data based on the class data, which classifies a digital image and/or classifies a target subject of the digital image. With the inclusion of the machine learning system **140** and the greedy cutout system **130**, the control system **420** is configured to generate actuator control data that allows for safer and more accurate control of the actuator system **430** of the vehicle as the greedy cutout system **130** provides the trained machine learning system **140** with certified robust defenses against adversarial patches. The actuator system **430** may include a braking system, a propulsion system, an engine, a drivetrain, a steering system, or any number and combination of actuators of the vehicle. The actuator system **430** is configured to control the vehicle so that the vehicle follows rules of the roads and avoids collisions based at least on the class data that is provided by the machine learning system **140** via the guidance of the greedy cutout system **130**.

(42) FIG. **6** is a diagram of the system **400** with respect to security technology **600** according to an example embodiment. As a non-limiting example, the security technology **600** includes at least a monitoring system, a control access system, a surveillance system, or any suitable type of security apparatus. For instance, as one example, FIG. **6** relates to security technology **600**, which is configured to physically control a locked state and an unlocked state of a lock of the door **610** and display an enhanced image/video on the display **620**. The sensor system **410** includes at least an image sensor that is configured to provide image/video data.

(43) The control system **420** is configured to obtain the image/video data from the sensor system **410**. The control system **420** is also configured to generate class data that classifies digital images via the machine learning system **140** with certified robustness against adversarial patches via the greedy cutout system **130**. In addition, the control system **420** is configured to generate actuator control data that allows for safer and more accurate control of the actuator system **430** by using class data, which is generated by the greedy cutout system **130** and the machine learning system **140**. The control system **420** is configured to display the source image, the class data of the machine learning system **140**, any data relating to the greedy cutout system **130** and/or the machine learning system **140**, any data relating to the computer vision application **150**, or any number and combination thereof on the display **620**.

(44) FIG. **7** is a diagram of the system **400** with respect to imaging technology **700** according to an example embodiment. As a non-limiting example, the imaging technology **700** includes a magnetic resonance imaging (MRI) apparatus, an x-ray imaging apparatus, an ultrasonic apparatus, a medical imaging apparatus, or any suitable type of imaging apparatus. In FIG. **7**, the sensor system **410** includes at least one imaging sensor. The control system **420** is configured to obtain image data from the sensor system **410**. The control system **420** is also configured to generate class data that classifies the image data via the machine learning system **140** with certified robustness against adversarial patches via the greedy cutout system **130**. In addition, the control system **420** is configured to provide more accurate medical information by using the class data, which is generated via the greedy cutout system **130** and the machine learning system **140**. In addition, the control system **420** is configured to display the any relevant data (e.g., any data relating to the greedy cutout system **130**, the machine learning system **140**, the computer vision application **150**, or any number and combination thereof) on the display **710**.

(45) As described in this disclosure, the embodiments provide several advantages and benefits. For example, the greedy cutout system **130** recognizes that training a classifier with standard cutout augmentation is not entirely effective for all two-masked images. Therefore, the greedy cutout system **130** considers the invariance of image classifiers to pixel masking with an objective to improve PatchCleanser certification. For example, in contrast to PatchCleanser, which uses Random Cutout augmentation, i.e., applying two masks of size 128×128 at random locations to 224×224 training images, the greedy cutout system **130** develops a systematic approach to find x' that are more empirically effective. In this regard, the greedy cutout system **130** provides a strategy to train the classifier with worst-case masked images. When compared with various other masking strategies (e.g., Random Cutout, Gutout or Cutout guided by Grad- CAM, Cutout with exhaustive



search, etc.), the greedy cutout system **130** has a training strategy, which improves certified robustness over other baselines across different datasets and architectures.

(46) FIG. **8A** and FIG. **8B** illustrate some reference examples to show some issues associated with classifying some masked versions of images while also further explaining advantages of the greedy cutout system **130** over random cutout augmentation (e.g., PatchCleanser). Specifically, FIG. **8A** illustrates an example of a digital image **800** that displays a bird **800A** on a branch of a tree. In addition, FIG. **8A** illustrates a digital image **810**, which shows a masked version of the digital image **800** in which a mask portion **810A** and another mask portion **810B** completely occlude the object of interest (i.e., the bird **800A**). As another example, FIG. **8A** also shows a digital image **820** that displays an ostrich **820A** on land. In addition, FIG. **8A** illustrate a digital image **810**, which shows a masked version of the digital image **820** in which a mask portion **810A** and another mask portion **810B** completely occlude the object of interest (i.e., the ostrich **820A**). Regarding digital image **810** and digital image **830**, certification is not possible because the mask portions completely occlude the object of interest (e.g., bird **800A** in digital image **810** and ostrich **820A** in digital image **830**).

(47) Meanwhile, FIG. **8B** illustrates a digital image **840** that displays a side view of a bird **840A** and a digital image **860** that displays a front view of a face of an ostrich **860A**. In addition, FIG. **8B** illustrates a digital image **850**, which shows a masked version of the digital image **840** in which a mask portion **850A** and another mask portion **850B** partially occlude the object of interest (i.e., the bird **840A**). Also, FIG. **8B** illustrates a digital image **870**, which shows a masked version of the digital image **860** in which a mask portion **870A** and another mask portion **870B** partially occlude the object of interest (i.e., the ostrich **860A**). Regarding digital image **850** and digital image **870**, these represent a few cases in which Random Cutout (e.g., PatchCleanser) fails to provide certification, whereas the greedy cutout system **130** succeeds in enabling certifying at least these cases of partial occlusion, as well as other cases of partial occlusion. In this regard, FIG. **8B** illustrate non-limiting examples that demonstrate a number of advantages of the greedy cutout system **130**, such as better invariance to pixel masking (compared to PatchCleanser) and improved certified robustness to adversarial patches.

(48) More specifically, with PatchCleanser, an image is certified if all 81 combinations of the masked versions of the image are classified correctly. However, during the certification process, there are some masked images (e.g. digital image **810** and digital image **820**), which are difficult for PatchCleanser to classify as some masks (e.g., masks **810A-810B** and masks **830A-830B**) completely occlude the object of interest (e.g., bird **800A** and ostrich **820A**) beyond recognition as shown in FIG. **8A**. It is not possible for Random Cutout trained classifiers to certify such images (e.g., image **800/810** and image **820/830**), as shown in FIG. **8A**, with the considered mask set. Besides, there has been visual observations that Random Cutout trained classifiers even misclassify some masked images (e.g. digital image **840** and digital image **870**) with preserved semantic clues, as shown in FIG. **8B**. Misclassification of such images (e.g., digital image **810**, digital image **830**, digital image **840**, digital image **870**, etc.) result in underestimating a classifier's certified robustness. In this regard, Random Cutout augmentation is not fully effective or invariant towards pixel masking and limits classifier potential to leverage the defense. Recognizing these misclassifications and inefficiencies of PatchCleanser, the greedy cutout system **130** explores training schemes that improve invariance to pixel masking and techniques to enhance a classifier's certified robustness. In addition, the greedy cutout system **130** is also advantageous in reducing the load and burden on the system **100** with a reduced training dataset compared to PatchCleanser. As discussed in this disclosure, the greedy cutout system **130** improves certification accuracy over PatchCleanser by a large margin and reduces the training burden, for example, via training the classifier with cutouts applied at worst- case regions, i.e., regions that produce the highest classification loss.

(49) That is, the above description is intended to be illustrative, and not restrictive, and provided in

the context of a particular application and its requirements. Those skilled in the art can appreciate from the foregoing description that the present invention may be implemented in a variety of forms, and that the various embodiments may be implemented alone or in combination. Therefore, while the embodiments of the present invention have been described in connection with particular examples thereof, the general principles defined herein may be applied to other embodiments and applications without departing from the spirit and scope of the described embodiments, and the true scope of the embodiments and/or methods of the present invention are not limited to the embodiments shown and described, since various modifications will become apparent to the skilled practitioner upon a study of the drawings, specification, and following claims. Additionally or alternatively, components and functionality may be separated or combined differently than in the manner of the various described embodiments, and may be described using different terminology. These and other variations, modifications, additions, and improvements may fall within the scope of the disclosure as defined in the claims that follow.

## Claims

1. A computer-implemented method for a certified robust defense against adversarial patch attacks, the method comprising: receiving a source image; generating a set of one-mask images using the source image and a first mask at a set of predetermined image regions, each one-mask image including the first mask at a different predetermined image region of the source image, the set of predetermined image regions collectively covering at least every pixel of the source image; generating, via a machine learning system, a set of predictions in response to receiving the set of one-mask images; selecting a particular one-mask image with a highest prediction loss from among the set of one-mask images; generating a set of two-mask images using the selected one-mask image and a second mask at the set of predetermined image regions, each two-mask image including the second mask at a different predetermined image region of the selected one-mask image; generating, via the machine learning system, another set of predictions in response to receiving the set of two-mask images; selecting a particular two-mask image with a highest prediction loss from among the set of two-mask images; generating a training dataset that includes the selected two-mask image; and training the machine learning system using the training dataset.
2. The computer-implemented method of claim 1, wherein: the set of two-mask images includes the selected two-mask image and non-selected two-mask images; and the training dataset does not include the non-selected two-mask images.
3. The computer-implemented method of claim 1, wherein the machine learning system includes a classifier that is trained to perform image classification using the training dataset.
4. The computer-implemented method of claim 1, wherein: the training dataset further includes the source image; and the source image does not include any adversarial pixels.
5. The computer-implemented method of claim 1, wherein the first mask covers a same size area as the second mask.
6. The computer-implemented method of claim 1, wherein: the source image is a digital image without any adversarial pixels; and the first mask is sized to cover a putative adversarial patch that is applied to no more than 3% of a total image area of the source image.
7. The computer-implemented method of claim 1, further comprising: receiving a digital image; employing the machine learning system to generate classification data upon classifying at least one object that is displayed in the digital image; and controlling an actuator based on the classification data.
8. A system comprising: at least one electronic processor; at least one non-transitory computer-readable medium in data communication with the at least one processor, the at least one computer-readable medium having computer readable data including instructions stored thereon that, when executed by the at least one processor, causes the at least one processor to perform a method for



defending against adversarial patch attacks with certified robustness, the method including: receiving a source image; generating a set of one-mask images using the source image and a first mask at a set of predetermined image regions, each one-mask image including the first mask at a different predetermined image region of the source image, the set of predetermined image regions collectively covering at least every pixel of the source image; generating, via a machine learning system, a set of predictions in response to receiving the set of one-mask images; selecting a particular one-mask image with a highest prediction loss from among the set of one-mask images; generating a set of two-mask images using the selected one-mask image and a second mask at the set of predetermined image regions, each two-mask image including the second mask at a different predetermined image region of the selected one-mask image; generating, via the machine learning system, another set of predictions in response to receiving the set of two-mask images; selecting a particular two-mask image with a highest prediction loss from among the set of two-mask images; generating a training dataset that includes the selected two-mask image; and training the machine learning system using the training dataset.

9. The system of claim 8, wherein: the set of two-mask images includes the selected two-mask image and non-selected two-mask images; and the training dataset does not include the non-selected two-mask images.

10. The system of claim 8, wherein the machine learning system includes a classifier that is trained to perform image classification using the training dataset.

11. The system of claim 8, wherein: the training dataset further includes the source image; and the source image does not include any adversarial pixels.

12. The system of claim 8, wherein the first mask covers a same size area as the second mask.

13. The system of claim 8, wherein: the source image is a digital image without any adversarial pixels; and the first mask is sized to cover a putative adversarial patch that is applied to no more than 3% of a total image area of the source image.

14. The system of claim 8, further comprising: a sensor that captures a digital image; and an actuator that is controlled based on classification data, wherein the machine learning system generates the classification data upon classifying at least one object that is displayed in the digital image.

15. A non-transitory computer readable medium having computer readable data including instructions stored thereon that, when executed by a processor, cause the processor to perform a method for defending against patch attacks, the method including: receiving a source image; generating a set of one-mask images using the source image and a first mask at a set of predetermined image regions, each one-mask image including the first mask at a different predetermined image region of the source image, the set of predetermined image regions collectively covering at least every pixel of the source image; generating, via a machine learning system, a set of predictions in response to receiving the set of one-mask images; selecting a particular one-mask image with a highest prediction loss from among the set of one-mask images; generating a set of two-mask images using the selected one-mask image and a second mask at the set of predetermined image regions, each two-mask image including the second mask at a different predetermined image region of the selected one-mask image; generating, via the machine learning system, another set of predictions in response to receiving the set of two-mask images; selecting a particular two-mask image with a highest prediction loss from among the set of two-mask images; generating a training dataset that includes the selected two-mask image; and training the machine learning system using the training dataset.

16. The non-transitory computer readable medium of claim 15, wherein: the set of two-mask images includes the selected two-mask image and non-selected two-mask images; and the training dataset does not include the non-selected two-mask images.

17. The non-transitory computer readable medium of claim 15, wherein the machine learning system includes a classifier that is trained to perform image classification using the training dataset.

18. The non-transitory computer readable medium of claim 15, wherein: the training dataset further includes the source image; and the source image does not include any adversarial pixels.
19. The non-transitory computer readable medium of claim 15, wherein the first mask covers a same size area as the second mask.
20. The non-transitory computer readable medium of claim 15, wherein: the source image is a digital image without any adversarial pixels; and the first mask is sized to cover a putative adversarial patch that is applied to no more than 3% of a total image area of the source image.
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